

2019-Ph-H1-Q4

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Summary of Question:

A car accelerates from rest on a flat road with **constant acceleration**.

You are asked to identify the correct **displacement-time (s–t)** and **acceleration-time (a–t)** graphs.

Answer:

For **constant acceleration**:

- The **acceleration-time graph** is a **horizontal line above the time axis** (constant positive value).
- The **displacement-time graph** is a **curve getting steeper** with time (because speed is increasing).

✅ **Correct answer: A**

Guidance for Students:

- Constant acceleration implies velocity is increasing uniformly, so displacement increases quadratically.
- The gradient of the s–t graph gives velocity; a steepening curve means increasing velocity.

Revision Tips:

- 📈 Know your standard graph shapes:
 - Constant velocity → straight sloped s–t line
 - Constant acceleration → curved s–t graph, straight a–t line
- 🧠 s–t gradient = velocity, v–t gradient = acceleration
- 🔄 Practise sketching motion graphs for different scenarios